I. Complete question 1-5 (3 points each).

Score	

1. Compute the limit $\lim\limits_{n\to\infty}\left(\frac{2}{n^2+1}+\frac{4}{n^2+2}+\cdots+\frac{2n}{n^2+n}\right)=$ ____1____.

2. Compute the limit $\lim\limits_{n\to\infty}\sqrt[n]{1^n+2^n+3^n}=$ ____3____.

3. If the limit $\lim\limits_{x\to 0}\left(\frac{e^{\frac{1}{x}}+1}{e^{\frac{1}{x}}-1}+\lambda\frac{\sin x}{|x|}\right)$ exists, then the constant $\lambda=$ _____.

4. If the function $\frac{x(x+1)}{1+\sqrt{x}}$ is the kth order infinitesimal of x as $x\to 0$, then the constant $k=$ _____.

5. If $\lim\limits_{x\to 1}\frac{x^2+ax+b}{x-1}=5$, then $a=$ ____3____, $b=$ ____-4____.

5. 间断

II. In question 6-10, please choose the correct answer (only one is correct) (3 points each).

Score	

6. The smallest positive period of the function $f(x)=\sin x+\cos 2x+\sin 3x$ is (B)

A. π B. 2π C. 3π D. 4π

7. If $f(x)=\lim\limits_{n\to\infty}\frac{1+x}{1+x^n}$ $(x>-1)$, then for $f(x)$ (C)

A. there is no discontinuous point. B. $x=0$ is a discontinuous point.

C. $x=1$ is a discontinuous point. D. $x=0,1$ are two discontinuous points.

8. If sequences $\{a_n\}$ and $\{b_n\}$ satisfy $\lim\limits_{n\to\infty}a_nb_n=0$, then (D)

A. If $\{a_n\}$ is divergent, then $\{b_n\}$ is divergent;

B. If $\{a_n\}$ is unbounded, then $\{b_n\}$ is bounded;

C. If $\{a_n\}$ is bounded, then $\{b_n\}$ is an infinitesimal;

D. If $\{1/a_n\}$ is an infinitesimal, then $\{b_n\}$ is also an infinitesimal.

9. Find the correct conclusion of the function $f(x)=\frac{1}{x}\sin\frac{1}{x}$ as $x\to 0^+$ (B)

A. Infinity. B. Unbounded function but not infinity.

C. Infinitesimal. D. Bounded function but not infinitesimal.

10. If the function

$$f(x)=\begin{cases}\frac{\ln(1+x^2)-\ln 2}{x^3-1} & x\neq 1\\ a & x=1\end{cases}$$

is continuous at $x=1$, then the constant a is (C)

A. $\frac{2}{3}$. B. $-\frac{2}{3}$. C. $\frac{1}{3}$. D. $-\frac{1}{3}$.

11. Prove $\lim\limits_{n\to\infty}\frac{3n+1}{2n+1}=\frac{3}{2}$ using the "$\varepsilon-N$" definition of a limit.

12. Prove $\lim\limits_{x\to -2}\frac{x^2-4}{x+2}=-4$ using the "$\varepsilon-\delta$" definition of a limit.

14. Compute

$$\lim_{x\to 0}\frac{e^{3x}-e^{2x}-e^x+1}{\ln\cos x}$$

15. If $x_1=\sqrt{3}$, $x_{n+1}=\sqrt{3+2x_n}$ $(n=1,2,\ldots)$, using the monotonic bounded principle to prove that the limit of sequence $\{x_n\}$ exists and find the limit.

16. Find $\lim\limits_{x\to\infty}\frac{1}{x}[x]$

17. Let

$$f(x) = \lim_{n\to\infty} \frac{(n-1)x}{nx^2+1},$$

find the discontinuous point of the function and determine the type of discontinuous point.

18. If $\lim\limits_{x\to 0} \frac{f(x)}{1-\cos x} = 4$, find

$$\lim_{x\to 0}\left(1+\frac{f(x)}{x}\right)^{\frac{1}{x}}.$$

15. Solution: We show that $\{x_n\}$ is bounded firstly.

Due to $0 < x_1 = \sqrt{3} < 3, 0 < x_2 = \sqrt{3+2x_1} < 3$. Suppose $0 < x_k < 3$, then

$0 < x_{k+1} = \sqrt{3+2x_k} < \sqrt{3+2\cdot 3} = 3$. According to mathematical induction, we get

$0 < x_n < 3 \quad n = 1, 2, \cdots$, This shows the sequence $\{x_n\}$ is bounded.

The monotonicity of $\{x_n\}$ is proved as follows.

Since $x_{n+1} - x_n = \sqrt{3+2x_n} - x_n = \dfrac{(3-x_n)(1+x_n)}{\sqrt{3+2x_n}+x_n} > 0$,

Thus $\{x_n\}$ is an increasing sequence.

According to the monotonic bounded principle, the limit of $\{x_n\}$ exists.

Assume $\lim\limits_{n\to\infty} x_n = a$, then $\lim\limits_{n\to\infty} x_n = \lim\limits_{n\to\infty}\sqrt{3+2x_n}$, i.e., $a = \sqrt{3+2a}$

$\Rightarrow a = 1(\text{delete})$ *or* $a = 3 \Rightarrow a = 3$.

Therefore, the sequence $\{x_n\}$ convergent and $\lim\limits_{n\to\infty} x_n = 3$. 8 points

I Completions (5 it

...ions (5 items, 3 points each)

1. 1 **2.** 3 **3.** -1 **4.** 1 **5.** $3, -4$

6. B 7.C 8. D 9. B 10. C

... and Proof (6 items, 8 points each)

11. Solution: $\forall \varepsilon > 0$,

in order to $\left|\frac{3n+1}{2n+1} - \frac{3}{2}\right| = \frac{1}{2(2n+1)} < \varepsilon$,

since $\frac{1}{2(2n+1)} < \frac{1}{4n}$,

We just need $\frac{1}{4n} < \varepsilon$, i.e., $n > \frac{1}{4\varepsilon}$.

Set $N = \left[\frac{1}{4\varepsilon}\right]$,

Then $\left|\frac{3n+1}{2n+1} - \frac{3}{2}\right| = \frac{1}{2(2n+1)} < \frac{1}{4n+1} < \varepsilon$ holds for $n > N$.

This shows $\lim_{n\to\infty} \frac{3n+1}{2n+1} = \frac{3}{2}$.

12. Solution: $\forall \varepsilon > 0$,

in order to $\left|\frac{x^2-4}{x+2} - (-4)\right| = \left|\frac{x^2+4x+4}{x+2}\right| = |x+2| < \varepsilon$, i.e., $|x-(-2)| < \varepsilon$.

Set $\delta = \varepsilon$,

Then $\left|\frac{x^2-4}{x+2} - (-4)\right| = \left|\frac{x^2+4x+4}{x+2}\right| = |x+2| < \varepsilon$ holds for $0 < |x-(-2)| < \delta$.

This shows $\lim_{x\to -2} \frac{x^2-4}{x+2} = -4$. 8 points

13. Solution: Set $t = -x \to +\infty$, then

$$\lim_{x\to-\infty} x\left(x + \sqrt{x^2+2}\right) = \lim_{t\to+\infty} -t\left(\sqrt{t^2+2} - t\right) = \lim_{t\to+\infty} \frac{-2t}{\sqrt{t^2+2}+t} = -1$$

14. Solution:

$$\lim_{x\to 0} \frac{e^{3x} - e^{2x} - e^x + 1}{\ln\cos x} = \lim_{x\to 0} \frac{(e^x-1)(e^{2x}-1)}{\ln\left[1+(\cos x - 1)\right]} = \lim_{x\to 0} \frac{x\cdot 2x}{-\frac{x^2}{2}} = -4$$

16. Solution: Due to $x - 1 < [x] \le x$

(i) $1 - \frac{1}{x} < \frac{1}{x}[x] \le \frac{x}{x} = 1$ *if* $x > 0$; (ii) $1 - \frac{1}{x} > \frac{1}{x}[x] \ge \frac{x}{x} = 1$ *if* $x < 0$;

Since $\lim_{x\to+\infty}\left(1 - \frac{1}{x}\right) = 1$ *and* $\lim_{x\to-\infty}\left(1 - \frac{1}{x}\right) = 1$,

According to the Sandwich Theorem, $\lim_{x\to+\infty} \frac{1}{x}[x] = 1$ and $\lim_{x\to-\infty} \frac{1}{x}[x] = 1$ hold.

Thus $\lim_{x\to\infty}\frac{1}{x}[x]=1$. 8 points

17. Solution: Since $f(x)=\lim_{n\to\infty}\frac{(n-1)x}{nx^2+1}=\lim_{n\to\infty}\frac{\frac{n-1}{n}x}{x^2+\frac{1}{n}}=\begin{cases}0 & x=0\\ \frac{1}{x} & x\neq 0\end{cases}$

Due to $\lim_{x\to 0} f(x)=\lim_{x\to 0}\frac{1}{x}=\infty\neq 0=f(0)$,

So $x=0$ is a infinite discontinuity point.

18. Solution: Due to $4=\lim_{x\to 0}\frac{f(x)}{1-\cos x}=\lim_{x\to 0}\frac{f(x)}{\frac{x^2}{2}}=2\lim_{x\to 0}\frac{\frac{f(x)}{x}}{x}$,

We obtain $\lim_{x\to 0}\frac{f(x)}{x}=0$, and $\lim_{x\to 0}\frac{f(x)}{x^2}=2$.

Thus $\lim_{x\to 0}\left(1+\frac{f(x)}{x}\right)^{\frac{1}{x}}=\lim_{x\to 0}\left(\left(1+\frac{f(x)}{x}\right)^{\frac{x}{f(x)}}\right)^{\frac{f(x)}{x^2}}=e^2$.

19. Solution: The concentration of salt is

$$C(t)=\frac{0.03\cdot 25t}{5000+25t}=\frac{0.75t}{5000+25t}\quad t\in[0,+\infty),$$

and $\lim_{t\to\infty}C(t)=\lim_{t\to\infty}\frac{0.75t}{5000+25t}=0.03=3\%$.

This shows the concentration of salt tends to be a 3% as $t\to\infty$ if the tank has sufficient large volume. 4points